

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – X

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Class:TYIT	Batch: B1
Date of Assignment:03/09/26	Date/Time of Submission:03/09/26

Aim

To perform data preparation and transformation using Tableau Prep Builder by connecting, cleaning, standardizing, merging, aggregating, filtering and exporting student assessment data.

Software Required

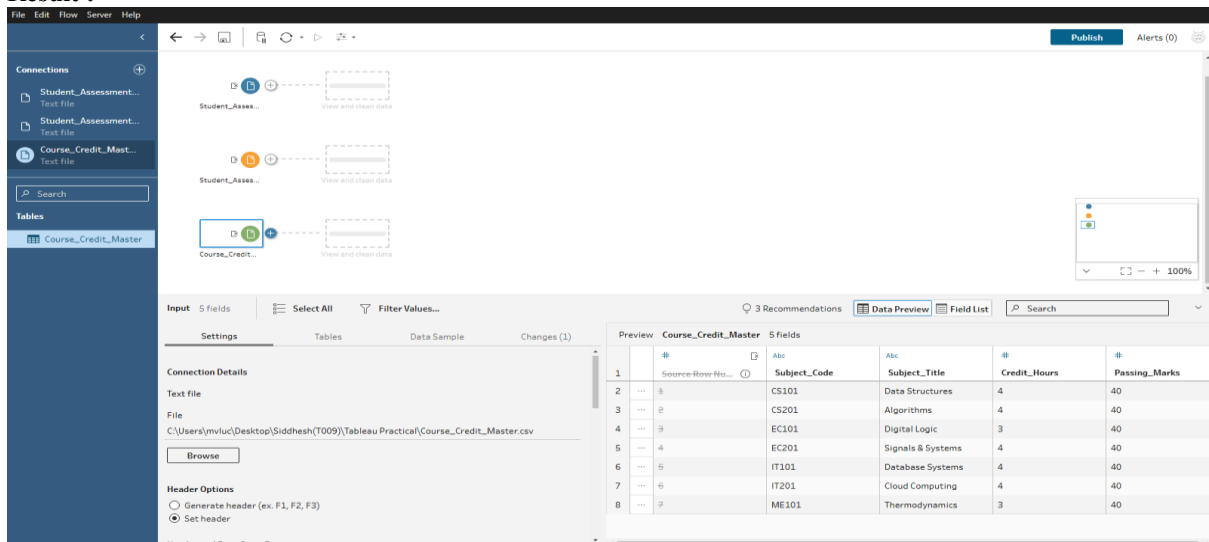
- Tableau Prep Builder
- Student_Assessments_Term1.csv
- Student_Assessments_Term2.csv
- Course_Credit_Master.csv

Procedure

A. Connect and Load Data

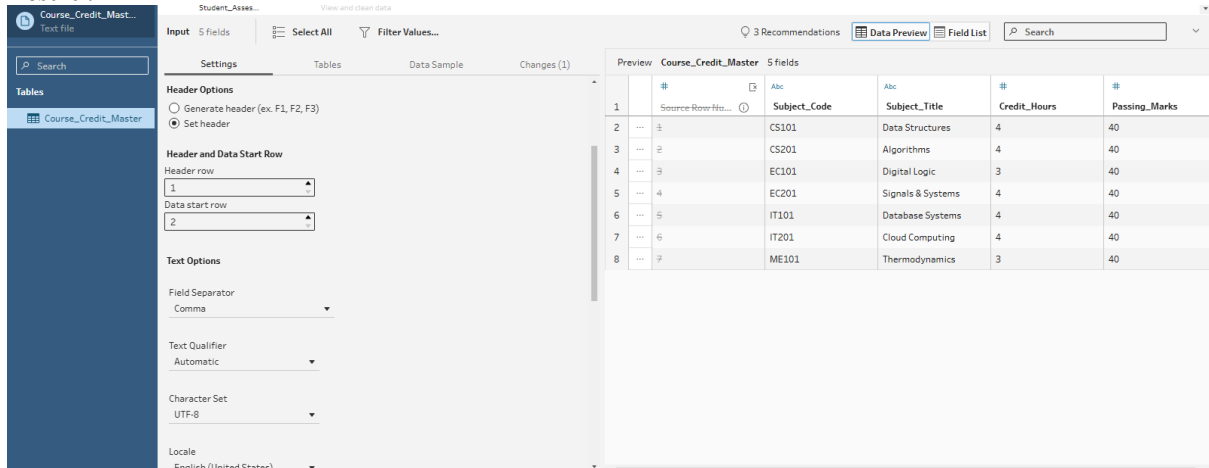
1. Open **Tableau Prep Builder** and connect the three CSV files using **Connections** → **+** → **Text File**.

Result :



2. Verify **Comma** as the field separator and select **First row contains headers**.

Result :



B. Clean and Profile Data

1. Add a **Clean Step** to Term 1 and inspect the **Profile Pane**.
2. Apply **Trim Spaces**, standardize student-name casing and change Exam_Date to **Date**.
3. Add a Clean Step to Term 2 and clean Full_Name and Assessment_Date.

Result :

The screenshot shows the Alteryx Designer interface. On the left, the 'Clean Step' menu is open, highlighting 'Trim Spaces'. The main workspace shows a data preview table with the following data:

#	Source Row #	Subject_Code	Subject_Title	Credit_Hours	Passing_Marks
1	3	CS101	Data Structures	4	40
2	4	CS201	Algorithms	4	40
3	5	EC101	Digital Logic	3	40
4	6	EC201	Signals & Systems	4	40
5	7	IT101	Database Systems	4	40
6	8	IT201	Cloud Computing	4	40
7	9	ME101	Thermodynamics	3	40

Clean Term 1 :

Student_Assessments_Term1 → student-name → (Trim Spaces)

The screenshot shows the Alteryx Designer interface. On the left, the 'Clean Step' menu is open, highlighting 'Trim Spaces'. The main workspace shows a data preview table with the following data:

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained
S101	Aarav Sharma	Comp Sci	Final	15-09-2025	CS101	35
S102	Aditya Rao	Computer Science	Midterm	20-11-2025	EC101	55
S103	Ananya Singh	CS		2025-09-15	IT101	
S104	Kavya Reddy	CSE		2025-11-20	ME101	75
S105	Priya VERMA	ECE		2025/09/15		
S106	Rahul Nair	Electronics		2025/11/20		95
S107	Rohan Gupta	Electronics Engg				
S108	Sneha Patel	Info Tech				
		Information Tech				
		IT				
		Mech Engg				

Student_Assessments_Term1 → student-name → (Make Uppercase)

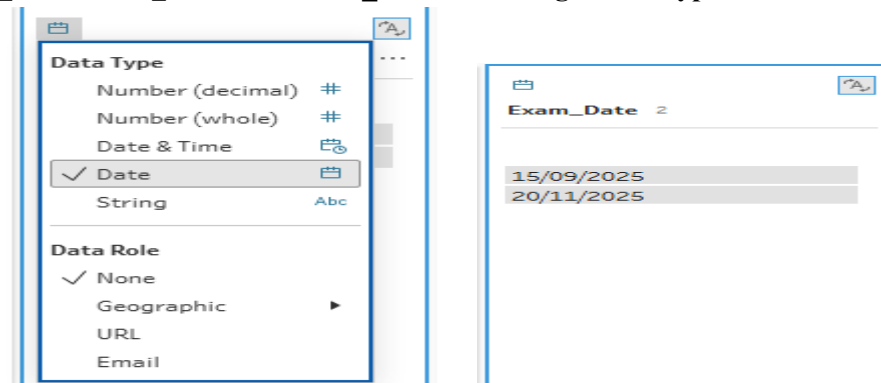
The screenshot shows the Alteryx Designer interface. On the left, the 'Clean Step' menu is open, and 'Make Uppercase' is selected under the 'Clean' sub-menu.

Clean 1 9 fields 16 rows

Select All Filter Values... Identify Duplicate Rows Rename Fields... Create Calculated Field... Search

Student_ID	Student_Name	Department	Assessment...	Exam_Date	Subject...	Marks_Obtai...
S101	AARAV SHARMA	Comp Sci	Final	15-09-2025	CS101	35
S102	ADITYA RAO	Computer Science	Midterm	20-11-2025	EC101	75
S103	ANANYA SINGH	CS		2025.09.15	IT101	75
S104	KAVYA REDDY	CSE		2025.11.20	ME101	95
S105	PRIYA VERMA	ECE		2025/09/15		
S106	RAHUL NAIR	Electronics		2025/11/20		
S107	ROHAN GUPTA	Electronics Engg				
S108	SNEHA PATEL	Info Tech				
		Information Tech				
		IT				
		Mech Engg				

Student_Assessments_Term1 → Exam_Date → Change Data Type Date



Clean Term 2 :

Clean 2 9 fields 14 rows

Select All Filter Values... Identify Duplicate Rows Rename Fields... Create Calculated Field... Search

Std_ID	Full_Name	Dept	Exam_Type	Assessment...	Course_Code	Score	Total_Pos
S101	Aarav Sharma	Computer Science	Final	10/02/2026	CS201	90	100
S102	Ananya Singh	CS	Midterm	25/04/2026	EC201	60	60
S103	Kavya Reddy	CSE			IT201	75	75
S104	Priya\NVerma	ECE				80	80
S105	Priya Verma	Electronics				82	82
S106	Rahul\Nair	Information Tech				94	94
S107	Rahul Nair	IT				95	95
S108	Rohan Gupta					98	98
S109	Sneha Patel					99	99
S110						100	100

Student_Assessments_Term2 → Full_Name → Clean → Trim Spaces.

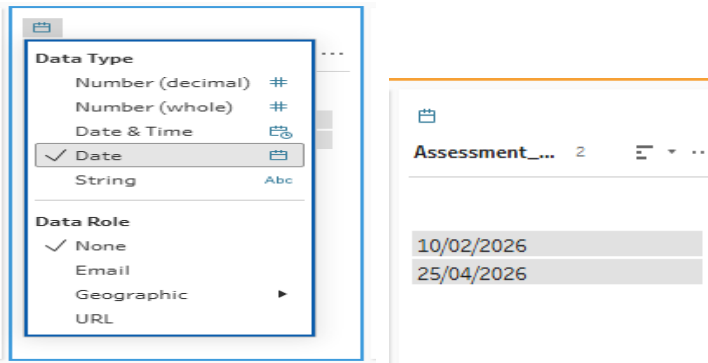
Full_Name

Filter Clean Group Values Split Values Identify Duplicate Rows View State Detail Summary Rename Field Duplicate Field Keep Only Field Create Calculated Field Hide Field Remove

Make Uppercase Make Lowercase Make Titlecase Remove Letters Remove Numbers Remove Punctuation Trim Spaces Remove Extra Spaces Remove All Spaces

Full_Name
Aarav Sharma
Ananya Singh
Kavya Reddy
Priya\NVerma
Priya Verma
Rahul\Nair
Rahul Nair
Rohan Gupta
Sneha Patel

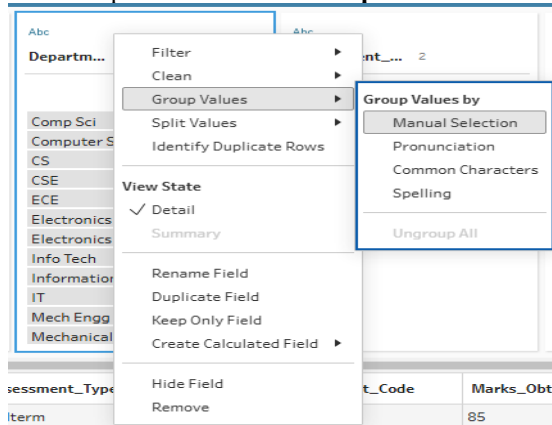
Student_Assessments_Term2 → Assessment_Date. → Change Data Type → Date



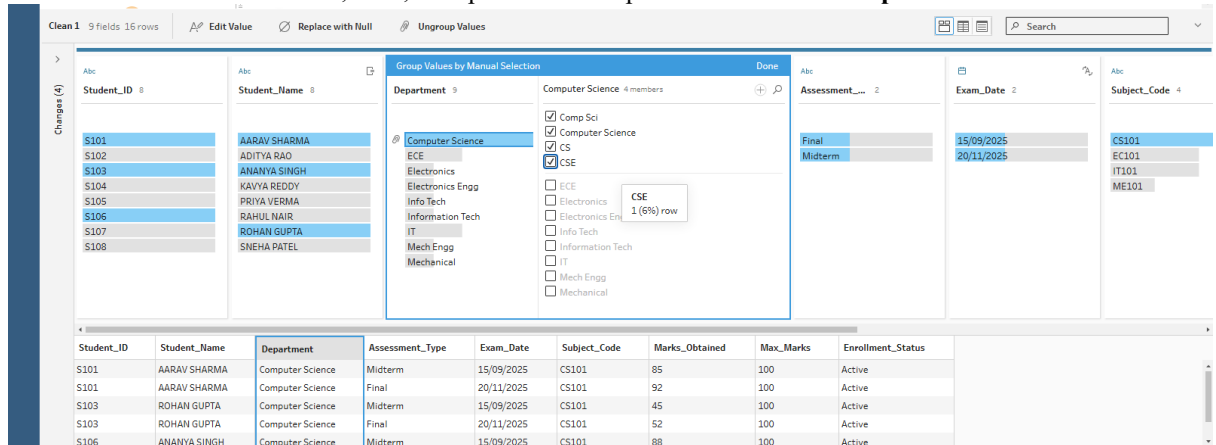
C. Group and Standardize Values

Clean 1 :

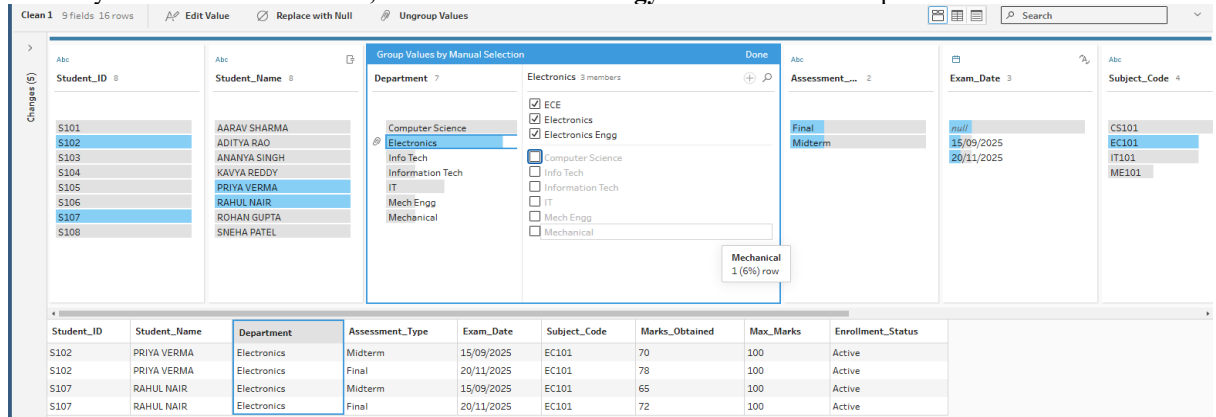
1. Select Department and use **Group Values**.



2. Combine variations such as CS, CSE, Comp Sci and Computer Science into **Computer Science**.



3. Similarly standardize **Electronics, Information Technology and Mechanical** departments.



4. Combine variations such as IT, Information Tech, and Info Tech into Information Technology (IT).

Clean 1 9 fields 16 rows

Group Values by Manual Selection

Department 5

Information Tech 3 members

Computer Science
Electronics
Information Tech
Mech Engg
Mechanical

Info Tech
Information Tech
IT

Computer Science
Electronics
Mech Engg
Mechanical

Student_ID 8

Student_Name 8

Assessment_Type 2

Exam_Date 2

Subject_Code 4

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S101	AARAV SHARMA	Information Tech	Midterm	15/09/2025	IT101	90	100	Active
S104	SNEHA PATEL	Information Tech	Final	20/11/2025	IT101	96	100	Active
S108	KAVYA REDDY	Information Tech	Midterm	15/09/2025	IT101	92	100	Active
S108	KAVYA REDDY	Information Tech	Final	20/11/2025	IT101	98	100	Active

5. Combine variations such as Mechanical and Mech Engg into Mechanical (ME).

Clean 1 9 fields 16 rows

Group Values by Manual Selection

Depta... 4

Mechanical 2 members

Computer Science
Electronics
Information Tech
Mechanical

Mech Engg
Mechanical

Mechanical 2 (13%) rows
2 (100%) highlighted

Student_ID 8

Student_Name 8

Assessment_Type 2

Exam_Date 3

Subject_Code 4

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S105	ADITYA RAO	Mechanical	Midterm	15/09/2025	ME101	35	100	Dropped
S105	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	40	100	Dropped

Clean 2 :

1. Combine variations such as CS, CSE, Comp Sci, and Computer Science into Computer Science (CS).

Clean 2 9 fields 14 rows

Group Values by Manual Selection

Dept 5

Computer Science 3 members

Computer Science
CS
CSE

ECE
Electronics
Information Tech
IT

Computer Science
CS
CSE

ECE
Electronics
Information Tech
IT

Std_ID 7

Full_Name 9

Exam_Type 2

Assessment_Date 2

Course_Code 3

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S106	Ananya Singh	Computer Science	Final	25/04/2026	CS201	96	100	Active
S101	Aarav Sharma	Computer Science	Midterm	10/02/2026	CS201	88	100	Active
S103	Rohan Gupta	Computer Science	Final	25/04/2026	CS201	60	100	Active
S101	Aarav Sharma	Computer Science	Final	25/04/2026	CS201	95	100	Active

2. Combine variations such as ECE, Electronics, and Electronics Engg into Electronics (EC).

Clean 2 9 fields 14 rows

Group Values by Manual Selection

Dept 4

Electronics 2 members

Computer Science
Electronics
Information Tech
IT

ECE
Electronics

Computer Science
Information Tech
IT

Std_ID 7

Full_Name 9

Exam_Type 2

Assessment_Date 2

Course_Code 3

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S107	Rahul/Nair	Electronics	Final	25/04/2026	EC201	75	100	Active
S102	Priya/Verma	Electronics	Final	25/04/2026	EC201	82	100	Active
S107	Rahul Nair	Electronics	Midterm	10/02/2026	EC201	70	100	Active
S102	Priya Verma	Electronics	Midterm	10/02/2026	EC201	75	100	Active

3. Combine variations such as **IT, Information Tech, and Info Tech** into **Information Technology (IT)**.

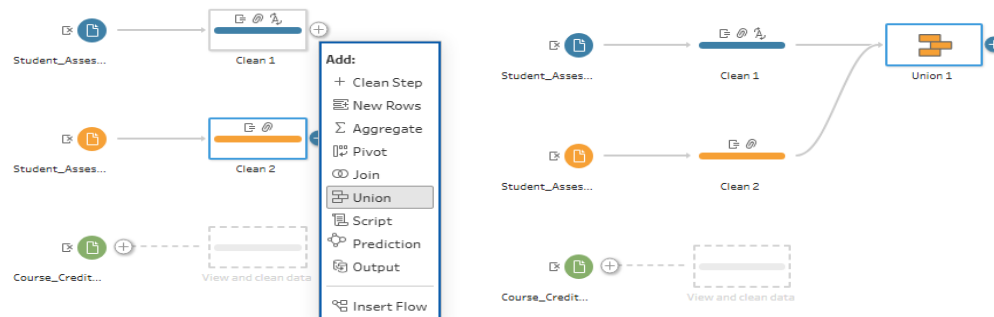
Clean 2 9 fields 14 rows | Edit Value | Replace with Null | Ungroup Values | Search

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S101	Aarav Sharma	Information Tech	Final	10/02/2026	IT201	98	100	Active
S102	Ananya Singh	Information Tech	Midterm	25/04/2026	EC201	95	100	Active
S103	Kavya Reddy	Information Tech	Final	25/04/2026	IT201	100	100	Active
S104	Priya Verma	Information Tech	Midterm	10/02/2026	IT201	94	100	Active
S106	Priya Verma	Information Tech	Final	10/02/2026	IT201	98	100	Active
S107	Rahul Nair	Information Tech	Midterm	25/04/2026	IT201	95	100	Active
S108	Rahul Nair	Information Tech	Final	25/04/2026	IT201	100	100	Active
S108	Rohan Gupta	Information Tech	Midterm	10/02/2026	IT201	94	100	Active
S108	Sneha Patel	Information Tech	Final	10/02/2026	IT201	98	100	Active

D. Merge Mismatched Fields

1. Create a **Union** between the cleaned Term 1 and Term 2 datasets.
2. Match the mismatched fields:
 - Std_ID → Student_ID
 - Full_Name → Student_Name
 - Dept → Department
 - Exam_Type → Assessment_Type
 - Assessment_Date → Exam_Date
 - Course_Code → Subject_Code
 - Score → Marks_Obtained
 - Total_Possible → Max_Marks

Result :



Union 1 17 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split Custom Split... Rename Field ... Search

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

14 Mismatching fields from 17 resulting fields.

Mismatched Fields

- Full_Name
- Student_Name
- Dept
- Exam_Type
- Assessment_Date
- Course_Code
- Score
- Total_Possible

Union Results Show only mismatched fields

Drop to merge fields

Student_Name Department Assessment_Type Exam_Date Subject_Code Marks_Obtained Max_Marks Full_Name Dept Exam_Type

Union 1 16 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split Custom Split... Rename Field ... Search

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

12 Mismatching fields from 16 resulting fields.

Mismatched Fields

- Dept
- Exam_Type
- Assessment_Date
- Course_Code
- Score
- Total_Possible
- Department
- Assessment_Type

Union Results Show only mismatched fields

Drop to merge fields

Department Assessment_Type Exam_Date Subject_Code Marks_Obtained Max_Marks Dept Exam_Type Assessment_Date Course_Code

Union 1 15 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split Custom Split... Rename Field ... Search

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

10 Mismatching fields from 15 resulting fields.

Mismatched Fields

- Exam_Type
- Assessment_Date
- Course_Code
- Score
- Total_Possible
- Assessment_Type
- Exam_Date
- Subject_Code

Union Results Show only mismatched fields

Drop to merge fields

Assessment_Type Exam_Date Subject_Code Marks_Obtained Max_Marks Exam_Type Assessment_Date Course_Code Score Total_Possible

Union 1 14 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Rename Field Create Calculated Field... Search

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

8 Mismatching fields from 14 resulting fields.

Mismatched Fields

- Assessment_Date
- Course_Code
- Score
- Total_Possible
- Exam_Date
- Subject_Code
- Marks_Obtained
- Max_Marks

Union Results Show only mismatched fields

Drop to merge fields

Exam_Date Subject_Code Marks_Obtained Max_Marks Assessment_Date Course_Code Score Total_Possible

Union 1 13 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split Custom Split... Rename Field ... Search

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

6 Mismatching fields from 13 resulting fields.

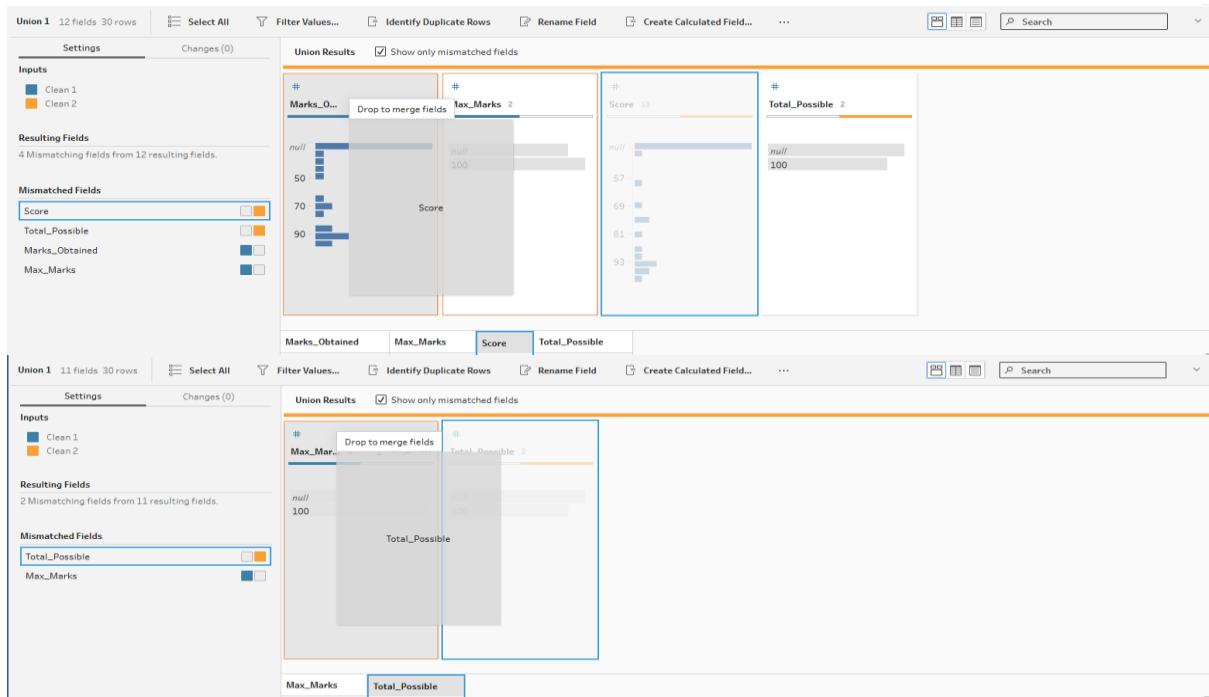
Mismatched Fields

- Course_Code
- Score
- Total_Possible
- Subject_Code
- Marks_Obtained
- Max_Marks

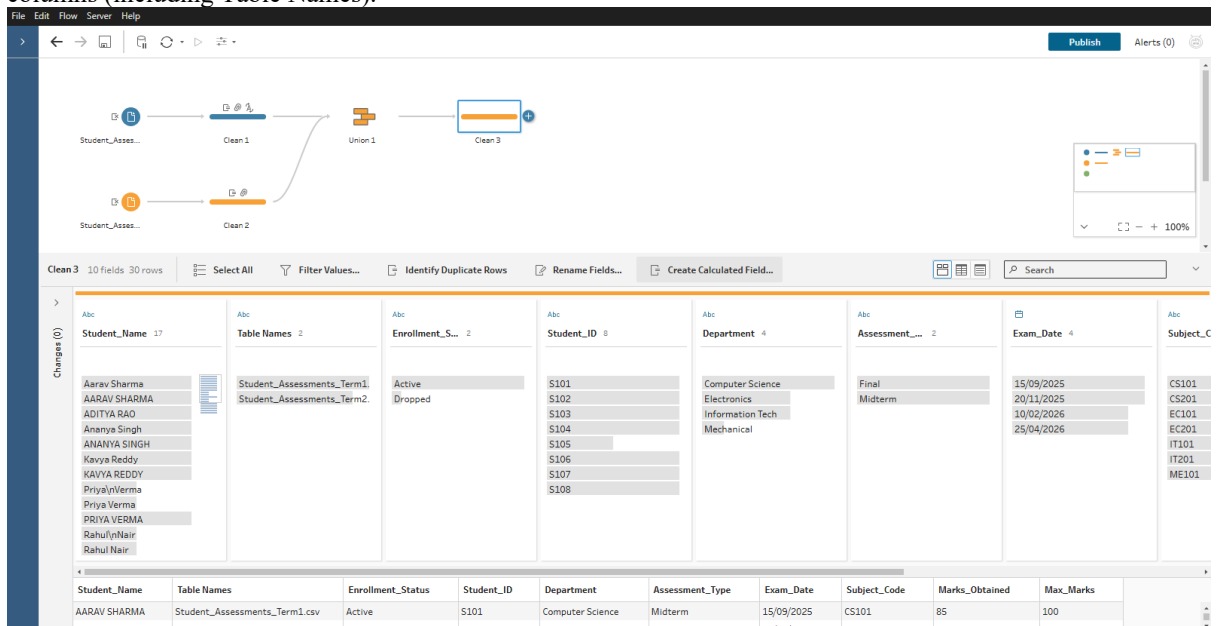
Union Results Show only mismatched fields

Drop to merge fields

Subject_Code Marks_Obtained Max_Marks Course_Code Score Total_Possible



Verify that the combined dataset now contains 30 total rows across 10 unified columns (including Table Names).



E. Perform Aggregation

1. Add an **Aggregate Step** after the Union.
2. Group the data by **Student_ID**.
3. Calculate **Total Marks, Average Marks and Assessment Count**.

Result :

Add Field

Field Name: Term

```
IF CONTAINS([Table Names], "Term1") THEN "Term 1"
ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"
ELSE "Other"
END
```

Calculation is valid

Apply Apply and Close

ABS(number)
Returns the absolute value of the given number.
Example: ABS(-7) = 7

File Edit View Server Help

Student_Asses... Clean 1 Union 1 Clean 3

Student_Asses... Clean 2

Changes [1]

Term 2

Student_Name 17

Table Names 2

Enrollment_Status 2

Student_ID 8

Department 4

Assessment_Type 2

Exam_Date

Term 1 AARAV SHARMA Student_Assessments_Term1.csv Active S101 Computer Science Midterm 15/09/2025 CS101 85 100

Term 1 AADAV SHARMA Student_Assessments_Term2.csv Active S101 Computer Science Midterm 20/11/2025 CS101 93 100

Aggregate 1 7 fields 17 rows

Settings Changes (0)

Additional Fields

Drag fields to aggregate or group them.

Search

Add All Remove All

GROUP Enrollment_Status

GROUP Exam_Date

GROUP Subject_Code

GROUP Table Names

SUM Number of Rows (Aggregated)

Grouped Fields

GROUP Student_Name 17

GROUP Student_ID 8

Aggregated Fields

AVG

marks_Obtain... 15

CNT

Assessment_Type 2

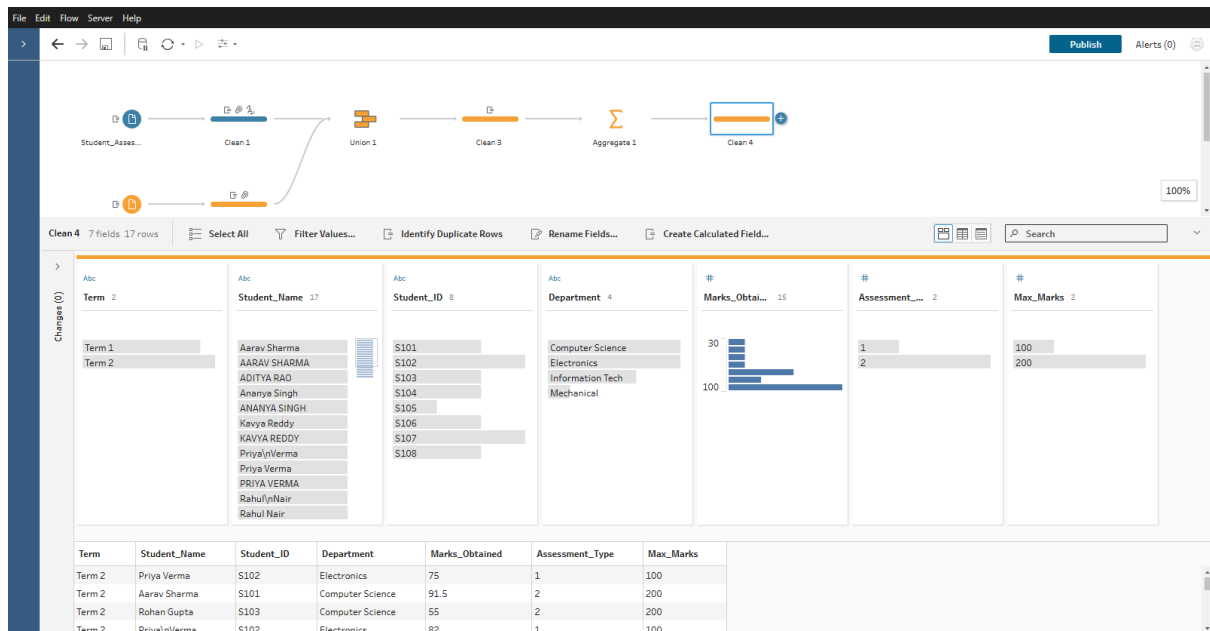
SUM

Max_Marks 2

Term Student_Name Student_ID Department Marks_Obtained Assessment_Type Max_Marks

Term 2 Priya Verma S102 Electronics 75 1 100

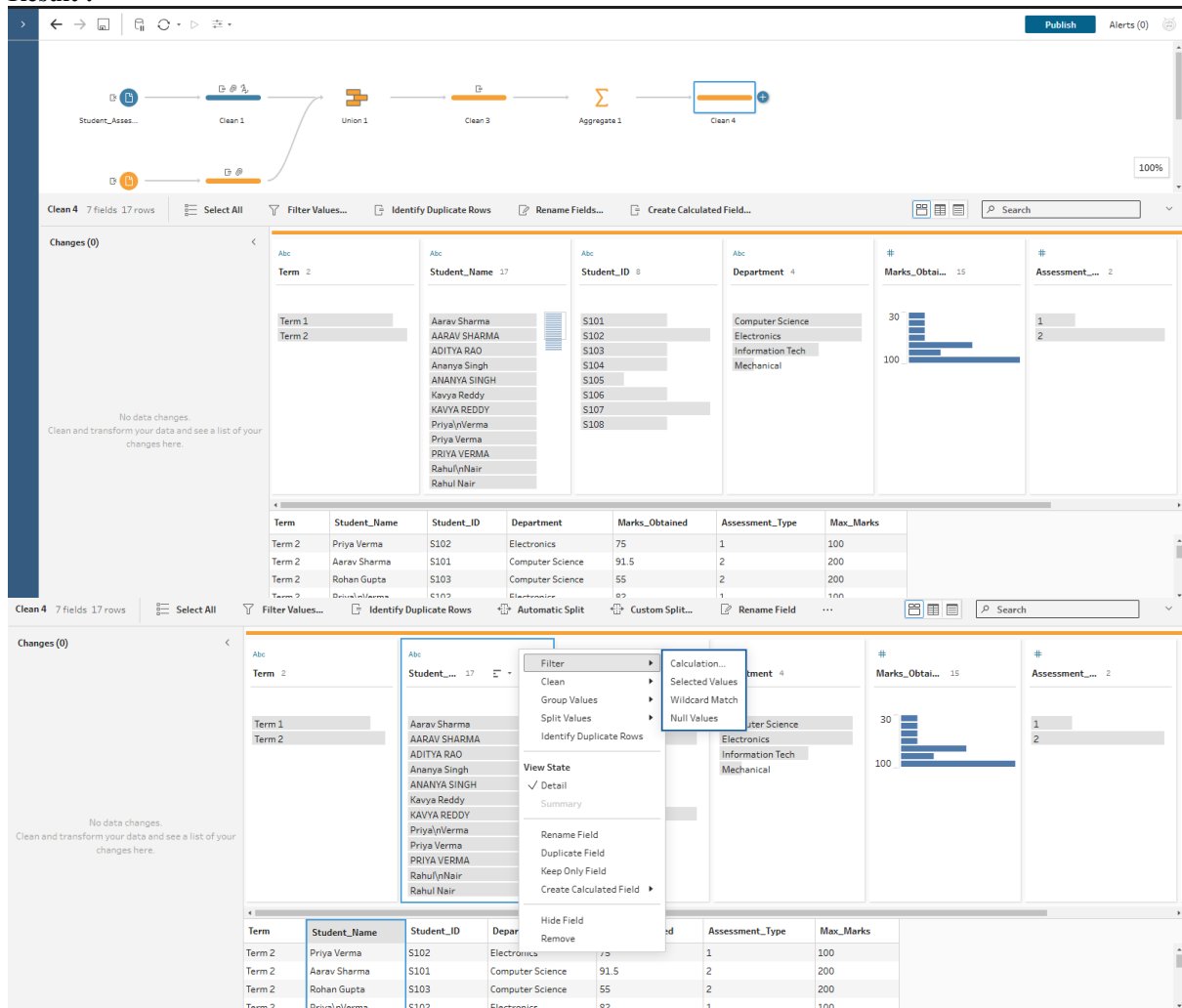
Term 2 Aarav Sharma S101 Computer Science 91.5 2 200



F. Apply Filters and Transformations

1. Filter Enrollment_Status and keep only **Active** students.
2. Create a Percentage calculated field using Marks Obtained and Maximum Marks.
3. Create Grade and Result_Status calculated fields based on Percentage.

Result :



Clean 4 7 fields 15 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split Custom Split... Rename Field ...

Changes (1)

Filter: Selected Values

Student_Name Exclude: 2 values

Filter: Selected Values

Student_Name 15

Keep Only Exclude 2

Search

AARAV SHARMA
ADITYA RAO
ANANYA SINGH
Kavya Reddy
KAVYA REDDY
Priya Verma
PRIYA VERMA
Rahul Nair
RAHUL NAIR
Rohan Gupta

☒ Aarav Sharma
☒ Priya Verma

Select All Clear All

Student_ID 8

S101
S102
S103
S104
S105
S106
S107
S108

Department 4

Computer Science
Electronics
Information Tech
Mechanical

Marks_Obtained 30
100

Edit Field

Field Name

Term_Percentage

$\text{ROUND}([Total_Marks_Obtained] / [Total_Max_Marks]) * 100, 2)$
 $\text{ROUND}([Score] / [Total_Possible]) * 100, 2)$

Calculation is valid

Edit with Tableau Agent

ABS(number)

Returns the absolute value of the given number.

Example: $\text{ABS}(-7) = 7$

Apply Apply and Close

Student_Ass... Clean 1 Union 1 Clean 9 Aggregate 1 Clean 10

Student_Ass... Clean 2

Clean 10 8 fields 8 rows

Select All Filter Values... Identify Duplicate Rows Rename Fields... Create Calculated Field...

Changes (4)

Term_Percentage 8

42
45
71
74
88
88.5
91
95

Total_Marks... 8

42
45
88
95
142
148
177
182

Term 2

Term 1
Term 2

Full_Name 6

Aarav Sharma
AARAV SHARMA
Aditya Rao
ANANYA SINGH
NEHA JOSHI
PRIYA VERMA

Student_ID 5

S101
S102
S105
S106
S108

#

Total_Ma... 2

100
200

Add Field

Field Name: Letter_Grade

```
IF [Term_Percentage] >= 85 THEN "A (Distinction)"
ELSEIF [Term_Percentage] >= 75 THEN "B (First Class)"
ELSEIF [Term_Percentage] >= 60 THEN "C (Second Class)"
ELSEIF [Term_Percentage] >= 40 THEN "D (Pass Class)"
ELSE "F (Fail / Re-appear)"
END
```

Calculation is valid

Create with Tableau Agent

Apply Apply and Close

Add Field

Field Name: Academic_Standing

```
IF [Term_Percentage] >= 60 THEN "Satisfactory"
ELSE "Academic Probation (<60%)"
END
```

ABS(number)

Returns the absolute value of the given number.

Example: ABS(-7) = 7

Clean 10 10 fields 8 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split Custom

Letter_Grade	Term_Percentage	Academic_Standing	Total_Marks
A (Distinction)	42	Academic Probation (<60%)	42
C (Second Class)	45	Satisfactory	45
D (Pass Class)	71		88
	74		95
	88		142
	88.5		148
	91		177
	95		182

G. Join and Output

1. Join the prepared assessment data with Course_Credit_Master.csv using Subject_Code.
2. Add an **Output Step** and select CSV or Tableau Hyper Extract.
3. Run the flow and verify the output.

Result :

Add:

- + Clean Step
- New Rows
- Aggregate
- Pivot
- Join
- Union
- Script
- Prediction
- Output**
- Insert Flow

Calculated Field...

Term 2

Full_Name 6

Output 10 fields

Save output to: File

Name: Student_Performance_Master_Cleaned

Location: C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Datasources

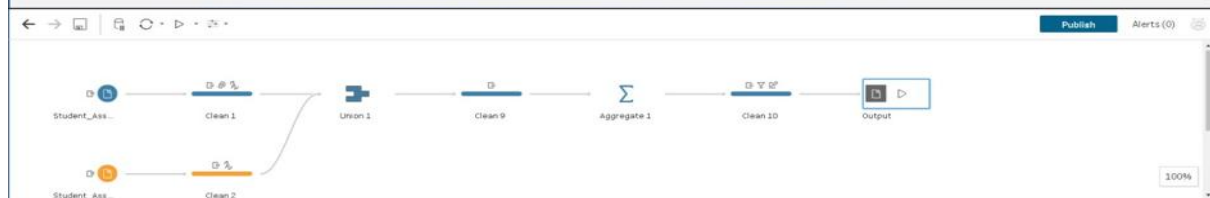
Output type: Tableau Data Extract (.hyper)

Write Options: Select an option to create or update your output table. Full refresh. Create table.

Run Flow

Save to Output.hyper

Letter_Grade	Term_Percentage	Academic_Standing	Total_Marks_Obtained	Term	Full_Name	Student_ID	Dept	Assessm
D (Pass Class)	45	Academic Probation (<60%)	45	Term 2	Aditya Rao	S105	Mechanical	1
C (Second Class)	74	Satisfactory	148	Term 1	PRIYA VERMA	S102	Electronics	2
A (Distinction)	88	Satisfactory	88	Term 2	Aarav Sharma	S101	CS	1
C (Second Class)	71	Satisfactory	142	Term 1	NEHA JOSHI	S108	Information Tech	2
D (Pass Class)	42	Academic Probation (<60%)	42	Term 2	Aditya Rao	S105	Mech Engg	1
A (Distinction)	91	Satisfactory	182	Term 1	ANANYA SINGH	S106	Computer Science	2
A (Distinction)	88.5	Satisfactory	177	Term 1	AARAV SHARMA	S101	Computer Science	2
A (Distinction)	95	Satisfactory	95	Term 2	Aarav Sharma	S101	Computer Science	1



Output 10 fields

Save output to: File

Name: Student_Performance_Master_Cleaned

Location: C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Datasources

Output type: Comma Separated Values (.csv)

Write Options: Select an option to create or update your output table. Full refresh. Create table.

Run Flow

Save to Student_Performance_Master_Cleaned.csv

Letter_Grade	Term_Percentage	Academic_Standing	Total_Marks_Obtained	Term	Full_Name	Student_ID	Dept	Assessm
D (Pass Class)	45	Academic Probation (<60%)	45	Term 2	Aditya Rao	S105	Mechanical	1
C (Second Class)	74	Satisfactory	148	Term 1	PRIYA VERMA	S102	Electronics	2
A (Distinction)	88	Satisfactory	88	Term 2	Aarav Sharma	S101	CS	1
C (Second Class)	71	Satisfactory	142	Term 1	NEHA JOSHI	S108	Information Tech	2
D (Pass Class)	42	Academic Probation (<60%)	42	Term 2	Aditya Rao	S105	Mech Engg	1
A (Distinction)	91	Satisfactory	182	Term 1	ANANYA SINGH	S106	Computer Science	2
A (Distinction)	88.5	Satisfactory	177	Term 1	AARAV SHARMA	S101	Computer Science	2
A (Distinction)	95	Satisfactory	95	Term 2	Aarav Sharma	S101	Computer Science	1

100%

Finished Running Flow

Student_Performance_Master_Cleaned.csv

Total time 00:02

Done

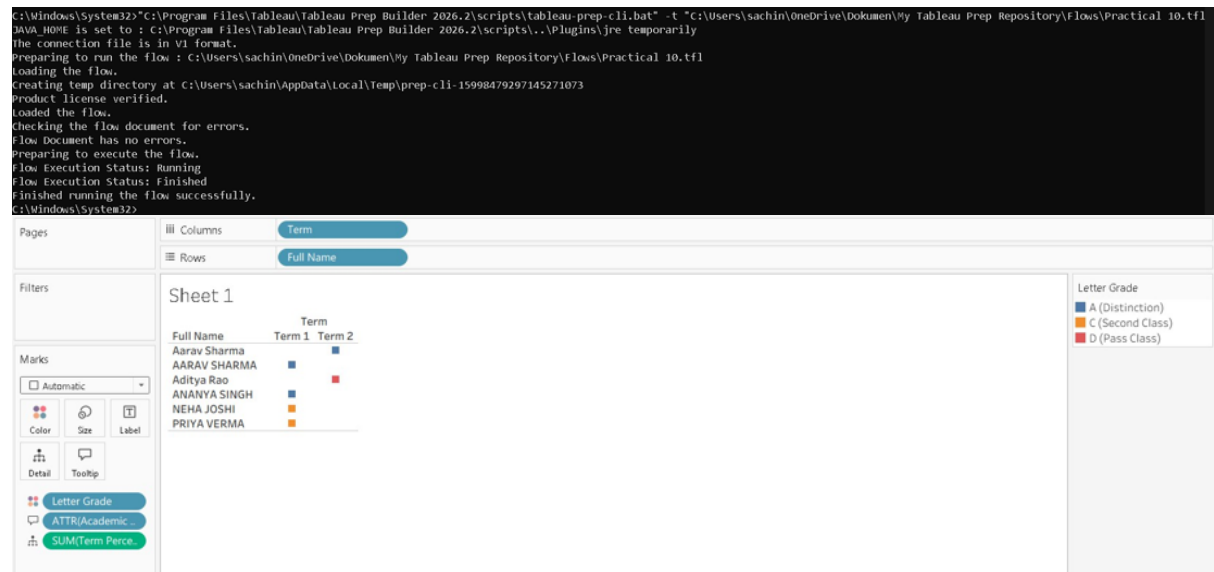
Term	Full_Name	Student_ID	Dept	Assessm
Term 2	Aditya Rao	S105	Mechanical	1
Term 1	PRIYA VERMA	S102	Electronics	2
Term 2	Aarav Sharma	S101	CS	1
Term 1	NEHA JOSHI	S108	Information Tech	2
Term 2	Aditya Rao	S105	Mech Engg	1
Term 1	ANANYA SINGH	S106	Computer Science	2
Term 1	AARAV SHARMA	S101	Computer Science	2
Term 2	Aarav Sharma	S101	Computer Science	1

```
C:\Windows\System32>dir "C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\*.tfl" /s
Volume in drive C is Windows-SSD
Volume Serial Number is D4E0-1C17
```

```
Directory of C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Flows
```

```
04-09-2026 12:43      28,734 Practical 10.tfl
04-09-2026 12:43      28,734 Student_Performance_Prep_Flow.tfl
                2 File(s)          57,468 bytes
```

```
Total Files Listed:
                2 File(s)          57,468 bytes
                0 Dir(s) 189,394,382,848 bytes free
```



Result

Successfully created a Tableau Prep data preparation flow to connect, clean, standardize, merge, aggregate, filter and transform student assessment data, and exported the final prepared dataset.